1(a)(i)	(halogens can) remove electron(s) (from a reactant)	**1**

1(d)(i)	**M1** $MgCl_2$ (is ionic and) has mobile ions **M2** $SiCl_4$ is covalent **AND** no charge carriers **OR** (no ions and) no delocalised electrons **OR** no mobile / moving ions	**2**

Examiner report missing
9701_w25_qp_24 Question 1

2(a)	**M1** Al_2O_3 is ionic **M2** P_4O_{10} is a (simple) molecule **M3** ionic bonds are stronger than intermolecular forces	**3**

Examiner report missing
9701_w25_qp_24 Question 2

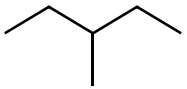
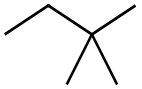
1(a)(i)	24 protons 28 neutrons 24 electrons	**1**
1(c)(i)	Cr / $Cr_2O_7^{2-}$ **AND** it gains electrons	**1**
1(c)(iv)	**M1** CO_2 has only instantaneous / temporary dipole–induced dipole forces **AND** methanal has permanent dipole–permanent dipole (and id-id) forces **M2** IMF's in methanal are stronger than IMF's in CO_2	**2**

Examiner report missing
9701_w25_qp_23 Question 1

2(a)(i)	delocalised electrons move (through the structure / between layers)	**1**
2(a)(ii)	P / S / Cl have no charge carriers **OR** (no ions and) no delocalised electrons **OR** no mobile / moving ions	**1**
2(b)	**M1** Si has (lots of) strong covalent bonds broken **M2** P / S / Cl (only) has weak Van der Waals' (forces) broken	**2**

Examiner report missing
9701_w25_qp_22 Question 2

3(a)	**M1** (family of molecules / compounds / series with) same general formula **OR** (family of molecules where) molecular formula differs (only) by (units of) $CH_{2(n)}$. **M2** similar chemical properties / reactions	**2**

3(d)(i)	**M1**  **M2** 	**2**

3(d)(ii)	**M1** B < A < hexane. **M2** hexane / unbranched has stronger / more IMFs **M3** hexane / unbranched requires most / more energy	**3**

Examiner report missing
9701_w25_qp_22 Question 3

1(a)(i)	24 protons 28 neutrons 24 electrons	**1**

1(c)(i)	Cr / $Cr_2O_7^{2-}$ **AND** it gains electrons	**1**

1(c)(iv)	**M1** CO_2 has only instantaneous / temporary dipole–induced dipole forces **AND** methanal has permanent dipole–permanent dipole (and id-id) forces **M2** IMF's in methanal are stronger than IMF's in CO_2	**2**

Examiner report missing
9701_w25_qp_21 Question 1

1(a)	<table><tr><td>*Mg*</td><td>*Si*</td><td>*P*</td></tr><tr><td>**M1** Giant metallic</td><td>**M2** Giant molecular</td><td>**M3** Simple molecular</td></tr></table>	**3**
1(b)	**M1** Na to A*l* have (delocalised) electrons that can move (through the structure / lattice) **AND** (P / S / C*l*) do not (have delocalised electrons / mobile charge carriers) **M2** Na to A*l* increasing number of delocalised electrons	**2**

Examiner report missing
9701_s25_qp_24 Question 1

6(a)(i)	simplest (whole number) ratio of atoms (of the elements) in a compound	**1**

6(b)	**M1** convert temp to 373 K (and use) **M2** correct rearrangement and use $7.15 = n8.31(4) \times T$ to find n **AND** Mr = 0.194 / n Mr = 84.1	**2**

Examiner report missing
9701_s25_qp_24 Question 6

2(a)	Na_2O	**1**

2(c)(i)	SO_3	**1**

2(c)(ii)	• false • Al_2O_3 melting point is lower than MgO • (but) charge on Al ion is greater than that on Mg Any 2 points = 1 mark , all 3 points = 2 marks	**2**

Examiner report missing
9701_s25_qp_23 Question 2

1(a)(i)	(both) giant molecular	**1**
1(a)(ii)	delocalised electrons move through the structure **OR** delocalised electrons move through / between the layers	**1**
1(a)(iii)	no delocalised electrons (to carry charge through the structure) **OR** <u>all</u> (4) outer electrons are used in bonding	**1**
1(b)(i)	$PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$	**1**
	$SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$	**1**
1(b)(ii)	**M1** phosphorus(V) chloride white / yellow solid **M2** silicon(IV) chloride colourless liquid	**2**
1(d)(i)	SO_2 –73 pd H_2O 0 H SiO_2 1600 C Na_2O 1132 I MgO 2852 I	**3**
1(d)(ii)	*Evaluate hypotheses based on the information candidates have given in the table* **M1** not enough information **M2** SO_2 and / or H_2O break intermolecular forces **OR** SO_2 and / or H_2O does not break covalent bonds **OR** only SiO_2 breaks covalent bonds	**2**

Examiner report missing
9701_s25_qp_22 Question 1

1(a)(i)	sodium – metallic **AND** sodium oxide – ionic	**1**

1(a)(ii)	delocalised electrons move (through the structure)	**1**

1(d)	*All three for two marks:* • false • comparison or statement of oxidation number • correct link to melting point from their comparison or statement	**2**

Examiner report missing
9701_s25_qp_21 Question 1

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	**2**
1(a)(iii)	ratio of moles of P : Cl = $\frac{30.4}{31.0}$: $\frac{69.6}{35.5}$ = 0.98 : 1.96 (= 1 : 2) molecular formula = P_2Cl_4	**2**
1(a)(iv)	(structure =) simple / molecular because low melting point / weak intermolecular forces (bonding =) covalent because hydrolysed	**2**

Examiner report missing
9701_m25_qp_22 Question 1

3(a)(i)	<table><tr><td>element</td><td>C (graphite)</td><td>Sn</td></tr><tr><td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr><tr><td>electrical conductivity</td><td>● good/ conductor</td><td>good</td></tr><tr><td>type of bonding</td><td>● covalent</td><td>metallic</td></tr><tr><td>type of structure</td><td>giant</td><td>● giant</td></tr></table>	**3**
3(a)(ii)	giant molecular	**1**
3(a)(iii)	its **delocalised** electrons are free to **move**	**1**
3(d)(i)	**M1** molecules CH_4 to PbH_4 have greater number of electrons / number of electrons increase	**1**
	M2 greater / stronger instantaneous dipole–induced forces / London forces / dispersion forces **OR** more energy required to overcome the instantaneous dipole–induced forces / London forces / dispersion forces	**1**
3(e)(i)	$Si + 2Cl_2 \rightarrow SiCl_4$	**1**
3(e)(iii)	**M1** SiO_2 is giant covalent **AND** $SiCl_4$ is simple covalent / simple molecular / molecular **M2** so need less **energy** to overcome IMF's in $SiCl_4$, compared to the **bonds** in SiO_2 **ora**	**2**

Question 3

(a) (i) This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

(ii) Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

(iii) Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

(b) (i) The most common errors were to give the formula for magnesium chloride as MgCl and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

(ii) Increases was correctly stated by many candidates.

(iii) Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, NH_4^+, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

(c) (i) Correct completion of the graph was shown by many candidates.

(ii) Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

(d) (i) There were many correct candidate answers stating an increase in the number of electrons, from CH_4 to PbH_4, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

(ii) The shape of a molecule of SiH_4 was well answered.

(e) (i) Many correct answers for the equation were present. A common error was to use HCl as a reactant, instead of Cl_2.

(ii) Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when SiCl_4 is added to water.

(iii) Credit was given to candidates for recognising that the structure and bonding of solid, SiO_2, was giant covalent and liquid SiCl_4 as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in SiO_2 but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid SiCl_4 molecules. Several answers compared the strength of the covalent bonds in both SiO_2 and SiCl_4 and many answers incorrectly stated that the structure of SiO_2 contained double bonds between the Si and O atoms.

(f) Many candidates gave the correct answer.

2(a)(i)	M1 $4Na + O_2 \rightarrow 2Na_2O$	**1**
	M2 $S + O_2 \rightarrow SO_2$	**1**

2(c)(i)	(molecules) H_2S to H_2Te / they have greater number of electrons	**1**
	stronger instantaneous dipole–induced dipole / London dispersion forces OR more energy required to overcome the instantaneous dipole–induced dipole / London dispersion forces	**1**

Question 2

(a) (i) The formula of sodium oxide was not well known. In equations where correct formulae were shown, a significant proportion were not balanced.

(ii) Candidate answers were rarely given in terms of an aluminium cation and an oxide anion. When ions were described, they were generally shown with the appropriate number of outer electrons; charges on the ions were sometimes omitted or incorrect.

(iii) Weak candidate answers did not fully address the question and explained why the maximum oxidation state varies because of the variable number of valence electrons.

(b) (i) Many candidate answers referred to $Mg(OH)_2$ with a pH of a slightly acidic solution rather than weakly alkaline. The product of the reaction of P_4O_{10} and H_2O was not well known, $P(OH)_5$ was a common incorrect answer.

(ii) This equation proved to be demanding. Common errors included equations which were incorrectly balanced or showed ammonia as a product of the reaction.

(iii) This question was answered well by candidates. Occasionally the same structure was shown twice, usually with propan-1-ol as both answers.

(iv) Excellent candidate answers were seen and these explained why there were fewer $H^+(aq)$ in solution in terms of the increase in strength of O-H bond due to the positive inductive effect of the ethyl group. Inappropriate use of the term 'electronegativity' instead of 'electron density' was seen. Common misconceptions included the production of OH^- ions when alcohols dissociate or that water fully dissociates or that alcohols contain OH^- ions.

(c) (i) Many candidate answers referred to H_2O although this was not relevant in **(c)(i)**. Weaker responses appreciated that there was an increase in strength of intermolecular forces but did not explain how this change arose. Many candidates gave answers in terms of an increase in size of the molecules or in terms of nuclear attraction, nuclear charge and shielding, which is incorrect. A common misconception described the increase in boiling point due to an increase in strength of permanent dipoles as the group is descended.

(ii) Identification of hydrogen bonds in H_2O was relatively common. The relevant strength of hydrogen bonds in water compared to those intermolecular forces present in the other hydrides was not always included. Incorrect reference to H_2S containing hydrogen bonds was seen and there was ambiguity in answers that referred to the relative strength of the O-H bonds rather than intermolecular hydrogen bonds.

3(a)(i)	(natural =)lightning	**1**
	(man-made =)internal combustion engines	**1**

3(d)	<table><tr><td>element</td><td>nitrogen</td><td>phosphorus</td></tr><tr><td>state and appearance</td><td>colourless gas</td><td>white solid</td></tr><tr><td>electrical conductivity</td><td>● poor</td><td>poor</td></tr><tr><td>type of bonding</td><td>● covalent</td><td>● covalent</td></tr><tr><td>type of structure</td><td>simple</td><td>● simple</td></tr></table>	**2**

3(e)	simple molecular (lattice structure)	**1**

Question 3

(a) (i) Many candidate answers referred to lightning as responsible for one natural occurrence of NO_2. A common incorrect answer was acid rain. Fewer candidate answers described the main source of man-made nitrogen oxides as the internal combustion engine. Lots of candidates incorrectly identified the source as catalytic converters or 'car exhaust' as the origin of nitrogen oxides.

(ii) Very few candidates gave appropriate species in the equation and if they did these equations were not always balanced correctly. Many missed the significance of the word 'direct' in the question and attempted to give an equation showing the role of NO_2 as a catalyst in the production of sulfur-based acid rain.

(iii) Candidates who were familiar with this area of the syllabus described how PAN forms from NO_2. Common incorrect descriptions included acid rain, water, oxygen, carbon monoxide, carbon dioxide or ozone reacting with NO_2.

(iv) Correctly balanced equations were common. Typical errors included incorrect formula of nitric acid or calcium nitrate or production of H_2 rather than H_2O.

(v) Some candidate answers described the production of a brown gas. Others focused on the production of a white solid even though the original compound is also a white solid.

(b) (i) Occasionally incorrect oxidation states were seen; –1 and +3 were common incorrect answers.

(ii) The majority of candidates identified aluminium as the correct species that is oxidised. The weakest answers described one of the two products of the reaction.

(iii) Excellent answers gave a full description of the formation of a dative bond due to the lone pair on the nitrogen atom accepting a proton. Weaker candidate descriptions only referred to the ability of ammonia to form a dative bond.

(iv) Identification of an appropriate shape for $Al(OH)_4^-$ was present in a significant proportion of answers.

(c) (i) Inclusion of state symbols for ionisation energy equations are essential. Some equations showed addition of an electron, other equations were incorrectly balanced.

(ii) A general increase in ionisation energy was shown by most candidates. Many answers showed a larger increase in energy for one of the ionisation processes, with the exact ionisation process often incorrectly identified. Weaker answers confused the trend in ionisation energy across a period with the trend of successive ionisation energies.

(d) Phosphorus was sometimes identified incorrectly with ionic bonding or as an ionic structure.

(e) Use of the term 'simple' to identify the structure of solids made up of discrete molecules was not well known and the structure of solid nitrogen was often referred to as 'giant'.

(f) (i) This question highlighted many misconceptions about hybridisation, sigma and pi bonds. These include

- a triple bond involves formation of two sigma bonds and one pi bond
- triple bonds are made when sigma and pi overlap
- hybridisation in P_2 identified as sp^2 or sp^3
- in P_2 overlap of s orbitals creates a sigma bond
- formation of sigma bonds by side-on overlap of orbitals and pi bonds by head-on overlap of orbitals.

(ii) Many candidate responses referred to the appropriate difference in bond strength. Some answers were incomplete, with no comparison of reactivity given. Some candidates misinterpreted the question and explained why they had different bond energy values.

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3(a)(ii)	giant molecular	**1**
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Question 3

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(b) (i) The most common errors were to give the formula for magnesium chloride as $MgCl$ and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

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(f) Many candidates gave the correct answer.

3(a)	$H_2(g) + ½O_2(g) \rightarrow H_2O(l)$ **M1** correct equation only **M2** correct state symbols based on this equation	**2**

3(c)(i)	giant (molecular and) covalent	**1**

Question 3

Candidates are urged to look carefully at the information provided in each item: here, this proved helpful and key to answering the ensuing physical chemistry questions.

(a) This was well answered by candidates. Standard states were required, that of water being (l).

(b) Common errors included incorrect stoichiometry or reverse parity for the answer. On the whole, the principle of Hess's Law seemed to be understood where answered.

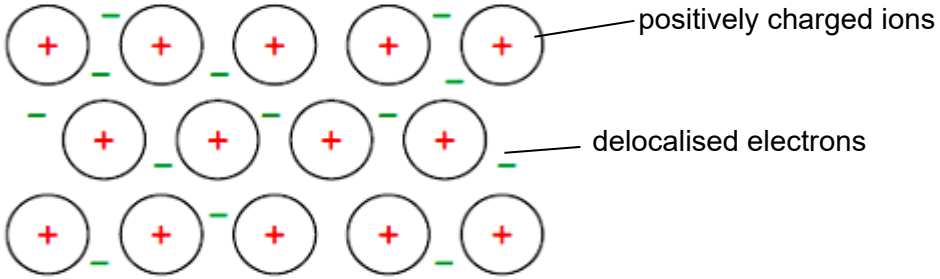
(c)(i) There were very many different answers here; it seemed that candidates were supposing that the reaction of 'metallic' boron and non-metallic carbon must result in a giant ionic lattice.

(ii) This was well answered by candidates who correctly used the A_r values of B and C. Some confusion was evident over the method of deciding the ratio of B : C. Answers without working were not credited, nor were answers trying to reason a formula from the ratio of putative charges of ions.

1(a)		**2**

Mg^{2+}	24	12	10
Al^{3+}	27	13	10

4 correct = 2 marks
2 or 3 correct = 1 mark
0 or 1 correct = 0 marks

1(c)	 labelled diagram shows: • regular arrangement of minimum of 2 × 2 circles containing + for positive ions • **AND** surrounded by (sea of) delocalised electrons	**1**
1(d)(i)	large amount of energy required to break bonds in giant covalent structure	**1**

Question 1

(a) The majority of answers were correct.

(b) Some excellent explanations were seen. Occasionally the aluminium ion was incorrectly described as larger than the magnesium ion. In general, the explanations focussed, at least in part, on the difference in nuclear attraction which arises as a result of the difference in nuclear charge and the same shielding effect of both species.

(c) Clearly labelled diagrams, which showed a regular arrangement of sodium cations surrounded by delocalised electrons, were seen in a significant proportion of answers. Poorer diagrams showed randomly placed positive ions or made no reference to 'delocalised' electrons. Occasionally, the positive symbols were described incorrectly as 'nuclei' or 'protons' or 'atoms'.

(d) (i) The majority identified the structure and bonding present in silicon. A much smaller proportion used this knowledge to explain why silicon had a high melting point in terms of the high energy requirement required to break many strong covalent bonds. A common misconception described intermolecular forces in a giant molecular structure.

(ii) Of those that annotated Fig.1.1, many correctly compared the melting points of P, S and C*l* with those of the three metals provided. Comparison of the melting points between P, S and C*l* was less well known.

(e) (i) This question proved to be demanding. The state of both sodium oxide and sulfur dioxide under room conditions was not well known. Misunderstanding of the term 'state of oxide' and oxidation number was seen on occasions. A common misconception described aqueous solutions of sodium oxide with a pH of 7.

(ii) Correct identification of this reaction as acid–base or neutralisation was common.

(iii) The product for the reaction of P_4O_{10} with an excess of water was well known. Equations were not always balanced.

(f) (i) The majority of answers demonstrated a good understanding of the term amphoteric.

(ii) There was some confusion with this reaction and the reaction of aluminium oxide and sodium hydroxide. The correct formula of the product was not well known.

2(a)(i)	hydrolysis	**1**
2(d)(i)	BN	**1**
2(d)(ii)	graphite	**1**

Question 2

(a) (i) This question was well answered.

(ii) The majority of answers gave a formula for the different Period 3 element when the question asked for a name. Common errors included phosphorus chloride, omitting the oxidation number, and phosphorus pentachloride, which is a solid.

(b) (i) The dot-and-cross diagram of SCl_2 was well answered. A common error included annotating the electrons around the S atom as 'x' in contradiction of the rubric.

(ii) The VSEPR theory was not well known when applied to a central S atom with two bonding pairs and two lone pairs of electrons in the outer shell. The majority of candidates focused on the two bonding pairs in SCl_2, stating that the shape was linear with a bond angle of 180°.

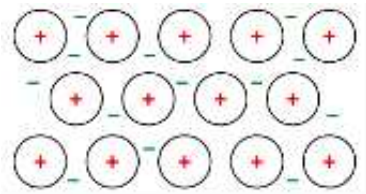
(c) (i) This question was well answered.

(ii) Most candidates gave a correct equation for the reaction of magnesium nitride with water. The most common error was an incorrect formula for magnesium nitride and magnesium hydroxide given as MgOH or MgO.

(iii) The marks were awarded to a minority of candidates. Very few candidates stated that hydroxide ions were present either from the dissociation of $Mg(OH)_2$ in solution, or from a reaction between ammonia and water.

(d) (i) The empirical formula of boron nitride was well answered with a common error of Br used to represent boron.

(ii) Many candidates gave the correct answer of graphite that has a structure similar to that of boron nitride. SiO_2 and graphene were common incorrect answers.

1(a)	 labelled diagram shows: • regular arrangement of positive ions • **AND** surrounded by / (sea of) delocalised electrons	**1**

1(b)(i)	$2Bi + 3H_2O \rightarrow Bi_2O_3 + 3H_2$	**1**

1(b)(ii)	giant **AND** has a high melting point	**1**
	ionic **AND** conducts electricity (only) when molten	**1**

Question 1

This was a broad-based question, encompassing ideas of physical, inorganic and organic chemistry. Candidates were adept with simple definitions, but application of ideas showed some deficit of understanding.

(a) Many diagrams were well presented, but a substantial number neglected to label the particles: mention of delocalisation of electrons was crucial. A sizable minority incorrectly identified the positive particles as protons or nuclei.

(b) (i) This question was mostly well answered, though errors of balancing and 'Bi_2' were often seen.

(ii) Although many candidates secured full credit here, common errors were not to link deduction and reasoning (e.g. giant *because* high melting point) and to offer contradictory statements (e.g. giant molecular and ionic bonding). In many cases, brevity served better. Some candidates appeared confused about the compound they were being asked to describe, Bi_2O_3, rather than bismuth itself.

(c) (i) This was answered well, though answers of +6 were commonly seen with respect to bismuth.

(ii) Many different answers were offered here, including $NaBH_4$ and O_2, whereas candidates needed to refer to a compound given in the equation.

(d) (i) This was generally well answered. It is preferable to refer to transfer of electrons when explaining the term oxidising agent.

(ii) Some contradictions were observed here, such as 'cold dilute … heat under reflux.'

(iii) Many candidates answered this well. Common errors included a switch of answers in the case of Tollens' reagent and neglecting to mention a precipitate or solid in the iodoform and DNPH reactions.

(iv) Many candidates tried unsuccessfully to include $NaBH_4$ in their answers. Candidates are reminded that even with the use of [H], there is a need for a stoichiometric balancing number. Use of CH_3COH is discouraged as a method of representing aldehydes.

(e) The initial stages of most calculations were good, but many candidates then forgot to multiply by 40 (to scale up to 1 dm^3 from 25 cm^3) or to multiply the concentration by 10 (1000 ÷ 100). In general, working needed to be shown more clearly, as some inventive methods could otherwise have given a seemingly correct answer.

2(a)	formula of chloride	● $NaCl$	● $MgCl_2$	● $SiCl_4$	**3**
	structure	giant	● giant	● simple	
	bonding of chloride	● ionic	● ionic	covalent	
	pH of solution formed	● 7	6.2	● 1–4	

Question 2

This was a holistic question based around the chemistry of chlorine and chlorides. Candidates are recommended to give brief answers to one-mark questions and to draw diagrams clearly.

(a) Common errors were found in the formulae of the chlorides, the structure of $SiCl_4$ and in the pH of a $NaCl$ solution (neutral so pH 7; candidates often seemed to think that: $NaCl + H_2O \rightarrow HCl + NaOH$).

(b) (i) Many variant and phonetic spellings of 'disproportionation' were observed.

(ii) There was an issue with this question in that water was not included as one of the products of the reaction, so full marks were awarded to all candidates to make sure no candidates were disadvantaged. This has been corrected in the published version of the paper.

(iii) This equation was rarely constructed correctly: where more valid attempts were made, common errors were either that $NaClO$ was given as the oxidation product or that the correct stoichiometric coefficients were omitted.

(iv) This was answered well by most candidates, although a number offered effervescence as an observation.

(c) (i) Octahedral was the most common answer for PCl_6^-, although commonly hexagonal or octagonal were seen (and not credited). A broad range of answers were given for PCl_4^+, with 'trigonal pyramidal' (missing the bi- prefix) and 'bipyramidal' (missing the base shape) among them.

(ii) **J** was often correctly identified as H_2O/water. Hydration or displacement were often given instead of hydrolysis or substitution for the type of reaction.

(d) (i) Some very loosely drawn diagrams were unable to be given full credit: curly arrows should come from a clear origin (the middle of a bond, or from pair of electrons, rather than a negative charge) and, in cases such as this, the arrowhead needs to have both 'blades' and point to a specific atom or the middle of a bond.

Many candidates also confused partial charges/dipoles with full formal charges.

(ii) Candidates were required to reference heating the ethanolic NaOH to gain full credit.

(iii) Many answers retained the C=C or arranged the 3C monomer linearly with no branching. These were both common and recurring errors in questions of this type.

1(a)(i)	total # 3p e^- 3 4 5	**1**
	total # unpaired e^- 3 2 1	**1**

1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	**1**
	(vigorous reaction with water suggests) **hydrolysis**	**1**

Question 1

(a) (i) Most candidates were able to complete the table to show the number of electrons in the 3p subshells of P, S and Cl, as 3, 4 and 5 respectively.

The total number of unpaired electrons in these elements proved challenging for some candidates, Common incorrect answers were 1, 0 and 1 respectively.

(ii) The equation to represent the first ionisation of P was well answered. A common error was to omit the state symbols of (g) throughout.

(ii) The question required candidates to choose one of three possible values, given in the question, to represent the first ionisation energy of S and to explain their choice.

Most candidates did not mention that S has a pair of electrons in a 3p **orbital**, which experience spin-pair repulsion. This results in S having an ionisation energy value of 1000 kJ mol^{-1}, which is lower than both P and Cl. The explanation required candidates to recall that whilst S has a nuclear charge higher than P and lower than Cl, its nuclear attraction to the outer electrons is the lowest of these three elements.

Many candidates chose a value between P and Cl and supported their choice with a incorrect statement that the 'nuclear attraction to the outer electrons' increased across the period.

(b) (i) The full electronic configuration of the P^{3-} ion was well answered as: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$.

(ii) The decrease in ionic radius from P^{3-} to S^{2-} and Cl^- was well answered. Many correct explanations included mention of an increased attraction of the outer electron to the nucleus resulting from an increased nuclear charge with each ionic species having the same shielding.

(c) Observations were required for three tests carried out on separate samples of NaCl(aq).

When Br_2(aq) is added to a sample no reaction occurs, and the colourless solution of NaCl(aq) turns orange (because of the added Br_2).

Most answers stated that the observation from the addition of concentrated H_2SO_4 to NaCl(aq) is misty/steamy fumes, not recognising that the NaCl is in aqueous solution (and not a solid). The solution remains colourless as there is no visible reaction.

A white precipitate is formed when $AgNO_3$(aq) is added to NaCl(aq).

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A small number of candidates gave a good explanation for this information and appreciated that the '*simple*' structure of $POCl_3$ was related to weak intermolecular forces of attractions and that the '*covalent*' nature of the compound is linked to the *hydrolysis* reaction with water.

(ii) The equation for the reaction of $POCl_3$ with water was generally quite well answered. Errors included an incorrect formula for H_3PO_4 and/or incorrect ratio between the water molecules and HCl product.

(iii) Most candidates were awarded full marks for the dot-and-cross diagram. A common error was the use of a dative bond between the P and O atoms.

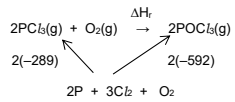
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Many answers mentioned standard conditions which was not a requirement.

(ii) The calculation of the bond energy of P=O (in PCl_3), using ΔH_f and bond energy (O_2) data, proved challenging for most candidates.

Use of incorrect energy cycles was the major problem and full marks were rarely awarded.

A correct energy cycle is:



ΔH_r = 2(–592) – 2(–289) = -606 kJmol^{-1}

bonds broken	bonds formed
~~6 P-Cl~~	~~6 P-Cl~~
1 O=O (496)	2 P=O

496 - 2P=O = -606
-2P=O = -1102
P=O = 551kJmol^{-1}

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**

3(d)(i)	$pV = nRT \therefore V = 8.31 \times 630 \div 100\,000 =$ **<u>0.0524</u>** (m^3)	**1**

3(d)(ii)	moles of gas = $12 \times 5.00 / 180$	**1**
	volume of gas = M1 × **(d)(i)** = **0.0175** (m^3)	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline I_2(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

1(a)(i)	decreases (from Si to S)	**1**

1(a)(ii)	**M1** Si giant covalent **AND** S is simple molecular **OR** Si giant covalent **AND** S has IMFs	**1**
	M2 Si involves breaking covalent bonds **AND** S involves breaking IMFs	**1**
	M3 more energy needed to break covalent bonds than IMFs **OR** covalent bonds are strong AND IMFs are weak	**1**

Question 1

(a) (i) The variation in atomic radius from silicon to sulfur was, in general, well known.

(ii) Many candidates appreciated that the differences in melting points were a consequence of the differences in structures. Identification of the types and relative strengths of the forces of attraction broken during melting were referred to less often. Common misconceptions included the presence of intermolecular forces in silicon or did not refer to intermolecular forces, neither in general terms, nor as instantaneous dipole induced dipole attractions in sulfur.

(b) (i) Most candidates correctly calculated the number of electrons in p subshells.

(ii) Many equations showed appropriate species. State symbols, essential in any equation to represent the first ionisation energy process, were included less often.

(iii) Candidates with an excellent knowledge of the trend in ionisation energy across a period identified the correct ionisation energy for phosphorus. They explained both the general trend and the effect of spin pair repulsion on a pair of electrons in a p orbital of sulfur compared to phosphorus. Weaker candidates either identified the value which represents a general trend in ionisation energy across the period and explained this trend or identified the correct value for ionisation energy with details that only explained why sulfur had a lower first ionisation energy than phosphorus.

(iv) This was answered well by most candidates.

(v) Appropriate description of the shape of an SCl_2 molecule as ‘bent’ or ‘v-shape’ was seen often. ‘Linear’ was a common incorrect answer.

1(a)(i)	total # 3p e^- 3 4 5	**1**
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1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	**1**
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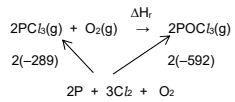
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(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

1(a)(i)	giant metallic	**1**

Question 1

(a) (i) A high proportion of candidates knew that copper is metallic. Many candidates omitted that the lattice is giant.

(ii) Diagrams showing the metallic bonding as an array of positive metal ions surrounded by delocalised electrons were varied, with many either omitting or not correctly labelling the positive ions and delocalised electrons.

(b) (i) Some candidates incorrectly referred to a mole of ^{12}C atoms or just an atom of ^{12}C.

(ii) Many correct answers were seen. Some candidates omitted the word 'average'.

(iii) The question was quite well answered. The major error included the use of 63 and 65 as the masses of the copper isotopes instead of the data given in the question.

(c) Very few candidates recognised that the ion with an abundance of 23% and an *m/e* of 31.5 was that of $^{63}Cu^{2+}$.

(d) (i) The equation was not well answered overall. Few candidates recognised that it was a redox reaction. Many answers contained electrons on one side of the equation and/or were completely unbalanced.

(ii) The identification of Cu^{2+} as the oxidising reagent in this reaction was generally not well known. Many candidates answered with just 'copper'.

(iii) The full electronic configuration of Cu^{2+} was frequently incorrect with the $4s^2$ electrons still present in many answers.

1(a)(i)	bonding in magnesium – metallic **AND** bonding in sodium chloride – ionic	**1**
1(a)(ii)	bonds in NaCl are stronger than bonds in Mg	**1**
1(b)(iii)	hydrogen bond	**1**
1(b)(v)	**M1** O is more electronegative than N **M2** two H-bonds per water molecule : 1 per ammonia molecule.	**2**

Question 1

(a) (i) It was generally well known about the bonding present in both species. Occasionally, candidates incorrectly described the bonding in magnesium as ionic.

(ii) The majority of candidates knew that the higher melting point seen for sodium chloride occurred because of stronger ionic bonds. Some candidates identified that there was a difference in bond strength between the two solids without using the information in the table to identify which had the stronger bond.

(iii) Many candidates were able to explain the difference in melting point in terms of the relative strength of intermolecular forces present. Stronger candidates identified the type of intermolecular forces involved and accounted for the difference in terms of the numbers of electrons present in molecules of sulfur and phosphorus. Some candidates incorrectly explained the difference in melting point being due to the difference in strength of covalent bonds.

(b) (i) This definition was generally well known. Some answers were incomplete or did not identify an atom as the appropriate species involved.

(ii) Many candidates knew that the outer electrons become more strongly attracted to the nucleus. Stronger candidates explained this effect in terms of the change in nuclear charge and the effect of shielding across the period.

(iii) It was generally well known that a hydrogen bond is the strongest intermolecular force present between ammonia molecules.

(iv) Stronger candidates were able to correctly show the role of the lone pair of electrons on the nitrogen atom in the formation of a hydrogen bond between ammonia molecules. Of those, some represented the attraction as a dative covalent bond with an arrow from the lone pair. Identification of appropriate partial charges in diagrams was seen more often. A common misconception showed two hydrogen bonds present between a pair of ammonia molecules to represent a dimer.

(v) The majority of candidates described the increased strength of a hydrogen bond in water and many accounted for the increased strength of the hydrogen bond by describing the difference in electronegativity between atoms of nitrogen and oxygen. A much smaller proportion of candidates also correctly referred to the different number of hydrogen bonds per molecule.

3(a)	**M1** *appearance* (colourless / yellow) liquid **M2** *structure and bonding* simple / molecular **AND** covalent	**2**

Question 3

(a) Very few candidates could state the appearance of $SiCl_4$ at room temperature; more were able to state the correct structure and bonding. The most common incorrect response was giant covalent.

(b) Most candidates correctly referred to the appearance and pH. Candidates often did not refer to the relevant observation and struggled to differentiate the behaviours of the two chlorides.

(c) (i) A large number of candidates stated the correct reagent. Many left out aqueous or dilute as the conditions.

(ii) Many of the candidates defined disproportionation; a small proportion were able to refer to the correct oxidation numbers. Several referred incorrectly to the +1 oxidation number for chlorine.

(d) (i) This was well answered by most candidates.

(ii) Most candidates correctly answered the initiation step. Many candidates found the propagation and termination steps more challenging.

(e) This was well answered by most students.

(f) Many candidates were able to give the correct formulae at m/e = 164.Very few candidates gave both of the correct options at m/e = 166.

1(a)(i)	power of an atom to attract electrons to itself	**1**

1(e)(i)	<table><tr><td>compound</td><td>O=C=O</td><td>O=S=O</td><td>S=C=S</td><td>S=C=O</td></tr><tr><td>overall dipole moment</td><td></td><td>✓</td><td></td><td>✓</td></tr></table>	**2**

1(e)(ii)	*conversion of units*	103000 Pa 127×10^{-6} m^3 423 K	**1**
	Use of $pV = (m / M_r)RT$	$M_r = \dfrac{0.284 \times 8.31 \times 423}{103000 \times 127 \times 10^{-6}}$	**1**
		M_r = 76.3 **AND** compound = CS_2	**1**

Question 1

This question tested ideas of atomic structure, periodicity and some quantitative chemistry. Knowledge of electronegativity was less secure and many Period 3 equations were incorrectly presented as redox reactions.

(a) (i) There were some very good answers here, although many candidates tried to answer in terms of ionisation or electron affinity.

(ii) Few candidates were awarded both marks as this required a balance of points to construct an argument. Spin-pair repulsion arguments were unnecessarily presented on a number of occasions. Many mentioned the distance of the outermost electron from the nucleus without any reference to its significance. Other answers included non-justified references to ionic radius or ionisation energy.

(b) (i) A whole array of different diagrams was seen, with only the ionic form (Li^+ and H^-) acceptable. All electrons, only four in total, were required to be shown.

(ii) Linear, trigonal planar and planar were popular incorrect answers.

(c) (i) The most common error here was the omission of state symbols from the elemental species. Many answers suggested that H_2 was being ionised.

(ii) This was well answered by many candidates.

(iii) Most candidates answered this well, although $3p^6$ was commonly seen.

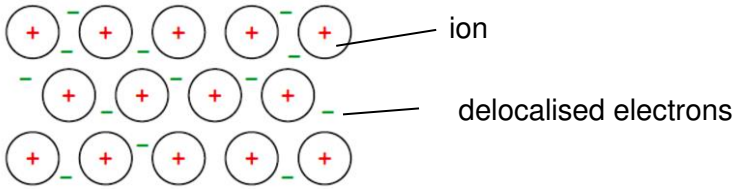
(d) (i) A large number of answers erroneously suggested H_2SO_4 as a product.

(ii) Both Na_2S and Na_2SO_4 appeared regularly, as did H_2. Candidates are advised to consider whether redox has occurred in these reactions.

(iii) Many incorrect versions for the formula of the carbonate were seen such as $Mg(CO_3)_2$ or $MgCO_2$. Many candidates incorrectly gave H_2 as a product from an acid–base reaction.

(e) (i) Many candidates gave the correct overall dipole for S=C=O but then went on to give the remaining three dipoles incorrectly.

(ii) Candidates frequently found it difficult to convert all of the physical quantities to consistent units. Many answers gave the identity as arsenic or some other element and clearly had not used Table 1.2 to identify the compound.

2(a)(i)	 **M1** diagram showing minimum of 4 particles (in total in 2 rows) • circles containing X^{n+} do not have to be labelled •  must be labelled as 'ion' OR empty circles / circles with X must be labelled + ion / positive ion / cation / X^{n+} **AND** with the circles surrounded by electrons shown as e^- / – **OR** little circles labelled electrons	**1**
	M2 label / legend showing **delocalised** electrons	**1**

2(a)(ii)	Mg has more delocalised e^- (than Na)	**1**

Question 2

This was an inorganic-chemistry question based around Group 2.

(a) (i) A labelled diagram showing the structure of a metal crystal was required here. Candidates are reminded of the subtle difference between delocalised and free electrons.

(ii) There were many good answers here although some responses continued to use free electrons instead of delocalised.

(b) A large proportion of candidates offered the reaction with steam ($\rightarrow MgO + H_2$) rather than that with cold water, although there were many good answers.

(c) Several answers gave examples of metals which would 'displace' Mg but not Ba. Water was a favourite reagent, with the explanation given as either difference in solubility or a hydrolysis type reaction. Such answers were not sufficiently creditworthy.

(d) (i) This question was answered quite well by candidates.

(ii) A good proportion of candidates understood what was needed, but many spoilt a good answer by stating that X = Ba. The most common error was not to include the stoichiometric factor of ½: such answers tended to have further inconsistency in the subtraction of the A_r of $1\times$ or $2 \times Cl$.